

Small-grid bounds for the no-five-on-a-sphere problem and an audit of canonical layer enumeration

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Abstract

Let $C(n)$ be the largest subset of $\{0, \dots, n-1\}^3$ with no five points on a common sphere or plane. We describe a search by coordinate layers and audit its implementation and retained results. Exact determinant checks certify configurations of sizes 8, 11, 14, 18 for $n = 3, 4, 5, 6$, respectively. For $n = 3, 4$, recorded SAT refutations with independently checked LRAT proofs exclude sizes 9, 12. For $n = 5, 6$, a corrected layer enumerator reports exhaustion of all 1,905 and 8,133 initial-layer orbits at target sizes 15, 19, respectively. These latter exclusions currently have evidence from one implementation, rather than independently reproduced decisions; they support the candidate exact values $C(5) = 14$ and $C(6) = 18$. We give the completeness argument for the layer reductions, correct an unsafe capacity prune, and distinguish visited-node counts from pruning counters. The public verification scripts reproduce the lower-bound checks and orbit inventories. The remaining task for the larger exclusions is an independent complete search or a separately checkable refutation.

Keywords. extremal combinatorics; integer grids; cospherical points; exact enumeration; reproducibility. **MSC 2020.** 05D99; 51M04; 68R05.

1 Problem and evidence

Write $G_n = \{0, \dots, n-1\}^3$. A five-element subset is *degenerate* when its points lie on a common sphere or plane. We study

$$C(n) = \max\{|S| : S \subseteq G_n, S \text{ contains no degenerate five-element subset}\}.$$

The inclusion of planes is essential: it is the convention used by the determinant verifier in the accompanying research repository [2]. A sphere-only problem would be a different extremal question. The no-five-on-a-sphere problem also appears among the construction tasks studied by PatternBoost [1].

Table 1 separates constructions, recorded proof checks and completed searches. A completed search log is useful evidence, but it does not by itself verify the search program’s mathematical reductions or implementation. In particular, we do not describe the $n = 5, 6$ exclusions as independently certified exact values. We make no claim of first priority for the values or constructions.

*Prepared and reviewed with AI research agents. The certificate checks, search records and remaining verification limits are distinguished in Section 4. No proof assistant was used.

n	verified construction	exclusion evidence
3	8	Size 9 excluded by recorded SAT/LRAT verification, with and without symmetry breaking.
4	11	Size 12 excluded by recorded SAT/LRAT verification, with and without symmetry breaking.
5	14	Size 15 absent from a completed corrected layer search; independent complete exclusion remains to be supplied.
6	18	Size 19 absent from a completed corrected layer search; independent complete exclusion remains to be supplied.

Table 1: Status of the small-grid results as of 28 September 2026. All construction checks use exact integer determinants.

2 Exact geometric predicates

For $p = (x, y, z)$ let

$$\ell(p) = (x, y, z, x^2 + y^2 + z^2, 1).$$

Five distinct points $p_1, \dots, p_5$ are degenerate if and only if

$$\det[\ell(p_i)]_{i=1}^5 = 0. \quad (1)$$

Indeed, linear dependence gives an equation $a(x^2 + y^2 + z^2) + bx + cy + dz + e = 0$. If $a = 0$, this defines a plane containing the points; if $a \neq 0$, completing squares gives a sphere. Its squared radius is nonnegative because it contains real points. The converse follows by substituting the equation of a plane or sphere. Thus every construction can be checked by evaluating all $\binom{|S|}{5}$ integer determinants.

Proposition 2.1. $C(3) \geq 8$, $C(4) \geq 11$, $C(5) \geq 14$ and $C(6) \geq 18$.

Proof. The accompanying coordinate files are listed in Section 5. The independent checker evaluates (1) on every five-element subset, as well as checking grid membership and distinctness. Every determinant is nonzero. $\square$

Each coordinate layer is a plane, so it contains at most four points of a feasible set. In particular $C(n) \leq 4n$. A four-element layer subset in a feasible set of size at least five must have linearly independent lifted rows. Within a layer this is equivalent to a nonzero determinant on the columns $(x, y, x^2 + y^2, 1)$. The excluded four-subsets are those lying on a circle or a line. Three collinear points together with a fourth point off their line are not excluded by this test.

The restriction follows because four dependent lifted rows remain dependent after a fifth row is added. Conversely, a nonzero layer determinant only validates the four-subset for extension; it does not certify any five-point set. For the 5×5 layer there are 11,824 such four-subsets out of $\binom{25}{4} = 12,650$; for the 6×6 layer there are 56,414 out of $\binom{36}{4} = 58,905$.

3 Layer enumeration

Fix the z coordinate and write a candidate as $S_0, \dots, S_{n-1}$ with $S_j \subseteq \{0, \dots, n-1\}^2$ and $|S_j| \leq 4$. The search places these subsets in increasing j , rejecting a placement as soon as it completes a degenerate five-subset.

3.1 Symmetry and the initial layer

The eight symmetries of a square act simultaneously on every layer. We choose one representative of each initial-layer orbit under this dihedral group D_4 . After placing more layers, a

candidate needs to be minimal only under the subgroup preserving every already placed layer. This stabilizer restriction identifies equivalent extensions without deleting all representatives of an extension orbit.

Reflection of the z axis allows $|S_0| \geq |S_{n-1}|$. At $n = 5$, target 15, an empty initial layer would force an empty final layer and leave capacity only 12, so the enumerator may start with nonempty layers. The inventory of valid initial subsets of sizes 1 through 4 has 1,905 dihedral orbits. At $n = 6$, target 19, initial and final layers of size at most one would leave capacity $1 + 4 \cdot 4 + 1 = 18$. Therefore one can reflect so that $|S_0| \geq 2$; subsets of sizes 2 through 4 have 8,133 dihedral orbits. The 8,139 orbits obtained when singletons are included differ by exactly six singleton orbits. These are separate inventories; the $n = 5$ count does not describe the $n = 6$ search.

3.2 Forbidden positions

For each four-subset of G_n , the program precomputes the positions in each layer that would complete a zero determinant. One bit mask per layer represents those positions. The $n = 5$ implementation stores five 32-bit masks per four-subset, using

$$\binom{125}{4} 5 \cdot 4 = 193,827,500 \text{ bytes.}$$

The $n = 6$ implementation uses six 64-bit masks, or

$$\binom{216}{4} 6 \cdot 8 = 4,233,656,160 \text{ bytes.}$$

These are table sizes, not total process memory. During extension, the forbidden-position mask is the union of the masks associated with the four-subsets already placed. Pair and triple compatibility checks handle five-subsets containing more than one new point. Incremental propagation avoids rebuilding contributions from four-subsets unaffected by the new layer.

3.3 A corrected capacity bound

Suppose layers $0, \dots, j - 1$ have been placed, with m points in total, and the target is K . For any future layer $r \geq j$, the other $n - j - 1$ unplaced layers can contribute at most four points each. Every completion must therefore satisfy

$$|S_r| \geq K - m - 4(n - j - 1). \quad (2)$$

The initial implementation incorrectly used $K - m - 4(n - r - 1)$, counting only layers after r as available capacity. For $r > j$ this is too large and can delete feasible branches. The corrected program uses (2). Its pair-compatibility pruning repeatedly removes positions with fewer than $\text{low} - 1$ compatible partners, where low is the right side of (2); every member of a feasible layer subset has at least this many partners, so this removal is sound.

Only the corrected runs are considered below. Source inspection and a successful feasible-instance control increase confidence, but neither substitutes for independent verification of the complete exclusion.

4 Retained results and verification limits

4.1 The smallest exclusions

The records for $n = 3$, target 9, and $n = 4$, target 12, report UNSAT and successful LRAT checking by a separate checker. Both symmetry-free and symmetry-reduced encodings were

checked. For example, the symmetry-free $n = 4$ record reports a formula with 14,932 variables and 224,365 clauses, and checker output **VERIFIED**. Together with Proposition 2.1, these recorded computations establish $C(3) = 8$ and $C(4) = 11$ subject to the correctness of the geometric encoding. The records, encoding source and construction checker are public. The large LRAT files are generated artifacts; this revision inspected the recorded checker results and did not rerun their checks.

4.2 Completed larger searches

The corrected $n = 5$, target-15 log records completion of all 1,905 distinct initial-layer representatives, with no partial-orbit marker and no solution. Its per-orbit node counts sum to 590,980,794. A target-14 control using the same source reports a solution. The source hash in the audit record agrees with the public enumerator.

The retained $n = 6$, target-19 log contains 8,175 completed-orbit lines covering all 8,133 distinct indices. Forty-two lines repeat already completed indices; their node counts agree. After deduplication, the visited-node total is 448,735,206,762. No orbit is marked partial and no solution is reported. The log’s `admiss` field counts admissibility-prune events, not visited nodes, and is not used as a search-size estimate. The public source is byte-identical to the source of this retained run.

An incremental and an earlier implementation agree on node counts for a recorded sample of thirty initial orbits. This checks a local optimization, not all reductions in the full search. Likewise, the built-in table self-test samples 200 four-subsets; its zero-mismatch report must not be described as an entrywise check of the entire table. The independent inventory checker reproduces the layer counts and orbit counts, but does not traverse the full extension search.

The larger exclusions consequently remain single-implementation computational results. The available $n = 6$ target-16 control succeeds; a target-18 control was deadline-limited and does not provide the stronger feasible-instance test. An attempted independent rerun was invalidated during this review: its subset iterator could repeat a branch indefinitely, its full-cache lookup lacked a termination condition, and its fully degenerate four-point case returned the wrong forbidden-position mask. Those runs supply no independent exclusion evidence. These defects concern the attempted cross-check, not the production enumerator described above. An independent full traversal, or a separately checked SAT refutation, is still needed before promoting the candidate equalities $C(5) = 14$ and $C(6) = 18$ to independently certified results.

5 Reproduction and data

All paths below are relative to the repository [2]. Lower-bound certificates are in `research/spikes/no-five-exact/certificates/`, in the files `n3_8.json`, `n4_11.json`, `n5_14_ls5x.json` and `n6_18_ls5x.json`. The checker `papers/no-five-exact/verify/check_small.py` independently verifies these coordinates and inventories the layer orbits using integer arithmetic. It does not claim to reproduce a full upper-bound search.

The search sources are `research/spikes/no-five-exact/layer_enum5.c` and `research/spikes/no-five-n6/layer_enum6.c`. Their command-line options, including target, orbit range and time limit, are documented in their source headers. A run with a deadline or an incomplete orbit range is not an exclusion proof. The SAT encoding and recorded solver/checker results are in `research/spikes/no-five-exact/`; see `encode_sat.py`, `solve_sat.py` and `records/`. The paper’s `REVIEW-NOTES.md` gives the source and log hashes used for this audit. Large tables, raw search logs and proof files are not embedded in the manuscript distribution; a reader reproducing an exclusion must regenerate and verify them.

6 Discussion

Coordinate layers reduce this extremal problem to a finite search with small local candidates. The useful result of the audit is a precise account of which reductions are sound, which quantities were counted and which decisions have independent verification. The explicit constructions are readily checkable; the recorded $n = 3, 4$ exclusions have separate proof-checker evidence; the larger searches provide candidate exact values with a remaining verification task.

No fitted formula for $C(n)$ is inferred from these data. In particular, $\lfloor (5n + 7)/2 \rfloor$ evaluates to 11, 13, 16, 18 at $n = 3, 4, 5, 6$, so it does not fit the small-grid values 8, 11, 14, 18. Extending the search to larger grids will require reducing the $\binom{n^3}{4}$ table cost while preserving a checkable account of every pruning rule.

References

- [1] F. Charton, J. S. Ellenberg, A. Z. Wagner and G. Williamson, *PatternBoost: constructions in mathematics with a little help from AI*, arXiv:2411.00566, 2024. <https://arxiv.org/abs/2411.00566>.
- [2] Hraness, *Algal Lab: extremal constructions and verification sources*, 2026. <https://github.com/hraness/algal-lab>. The exact source and evidence identifiers used here are recorded alongside this manuscript.